

Slide 1: The Problem & Opportunity
The Problem
Resource Waste: Unchecked tank and dam overflows bleed millions of gallons of clean water.
Infrastructure Damage: Running completely dry destroys expensive public pumping machinery.
Delayed Operations: Manual monitoring creates critical blind spots in remote locations.
The Opportunity
Modernizing Storage: Transitioning infrastructure from passive containers



to smart, reactive assets.



Slide 2: The HydroGuard Solution
Instant Automation
Microcontroller Intel: An Arduino core monitors fluid metrics and acts without human lag.
Visual Status: Multi-tier LEDs show live safe, transition, and warning zones instantly.
Auto-Correction: Electronic relays shut down pumps or open drainage valves before disaster strikes.
Real-World Uses
City Utilities: Municipal drinking towers.
Agriculture: Automated irrigation



reservoirs. Industrial Safety: Structural dam overflow control.



Slide 3: What Makes This System Unique
Wireless Field Overrides
Infrared Control: Operators use handheld remotes to test or mute systems from a distance.
Enhanced Safety: Eliminates the hazard of climbing high platforms or opening heavy hatches.
Bulletproof Reliability
Zero-Screen Diagnostics: Visual signals allow immediate inspection without using laptops in rough weather.
Dual Hardware Bypass: Integrated buttons ensure physical manual control remains if sensor logic freezes

